

Your First Program

The C++ 20/23/26 Masterclass

What We Do: .

- Entry Point
- Build and Run on The Command Line
- Output to the Screen
- Debugging through Qt Creator and Visual Studio (IDE)

THE FILES

- `CMakeLists.txt` - the build instructions for CMake, which we use to compile and run our program
- `main.cpp` - the main program file, where we write our code

MAIN.CPP

The Entry Point

```
#include <print>

int main() {

    std::println("Hello there!");
    std::println("Welcome to your very first C++ program.");

    return 0;
}
```

The Build Instructions

```
cmake_minimum_required(VERSION 3.20)

# Project name
project(rooster)

# Set C++ standard to C++23
set(CMAKE_CXX_STANDARD 23)
set(CMAKE_CXX_STANDARD_REQUIRED ON)

# Create executable
add_executable(rooster main.cpp)

# For multi-file projects with headers and separate .cpp files
# add_executable(rooster main.cpp dog.cpp dog.h cat.cpp cat.h)
```

RUN IT

Build and run

Same three commands every lecture in this course:

```
cmake -B build -G Ninja .  
cmake --build build  
./build/rooster
```

RUN IT

Run the Executable

```
build/rooster
```

COMMENTS

Two kinds of comments

```
// A one-line comment - the compiler ignores everything after //
```

```
/*  
  A block comment.  
  Can span multiple lines - handy for a longer explanation.  
*/
```

Rule of thumb: a comment earns its place by explaining **why**, not by narrating what the code obviously already says.